

VOYAGEOS

Autonomous Travel Planning & Re-Planning Agent

"Most travel apps plan your trip. VoyageOS manages it."

Problem-Solution Summary | Agentic AI Hackathon — Tech Zephyr 4.0, IIT Bhubaneswar

Team Versetify • github.com/Priyank-14/voyage-os

THE PROBLEM

Static itineraries break the moment a real trip begins.



Weather disrupts plans

Sudden monsoon downpours cancel white-water rafting or mountain treks.



Venues close unexpectedly

Historic monuments or ashrams shut down for unannounced maintenance.



Transit falls apart

Highway landslides or city traffic double transit times.



Budgets get blown

Incidental expenses threaten hard financial limits.

Bottom line: this is an adaptive multi-criteria decision-making problem — not a creative text-generation problem.

THE SOLUTION

VoyageOS: a stateful travel operations agent

It doesn't just generate a plan once — it perceives disruptions, evaluates them against hard constraints, and adapts the itinerary autonomously, preserving everything unaffected.

1

Continuous Perception

Tracks weather, venue status, and transit conditions in real time — not a one-shot prompt.

2

Deterministic Decisions

Hard constraints (e.g. rain $\geq$ 70%, budget ceilings) run as policy code, eliminating hallucinated judgment calls.

3

Minimal Replanning

Only the disrupted activity slot changes. Everything else in the itinerary stays exactly as booked.

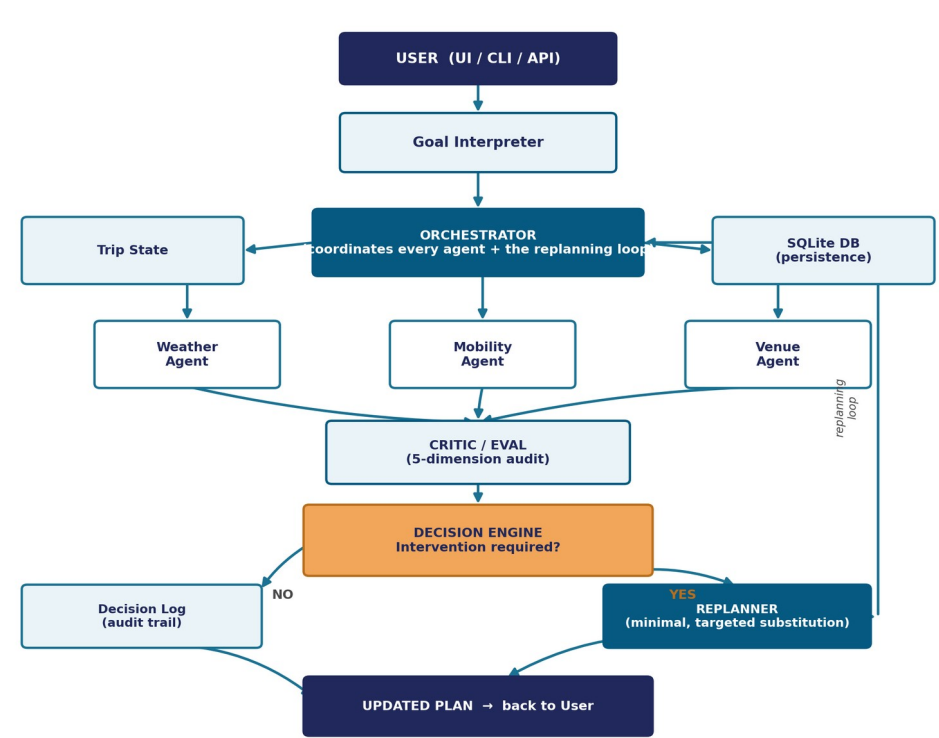
4

Auditable & Explainable

Every adaptation is logged with timestamp, rationale, and score breakdown — never an unexplained change.

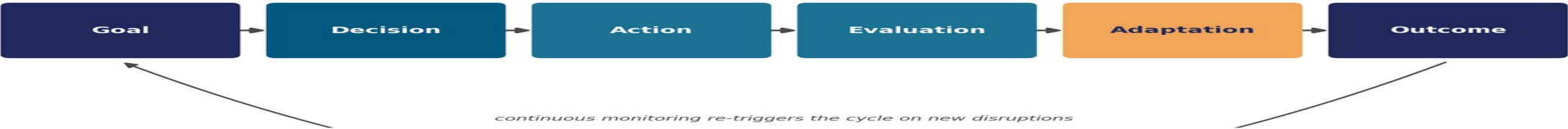
SYSTEM ARCHITECTURE

Full pipeline: sensing agents, deterministic decisions, and the replanning loop



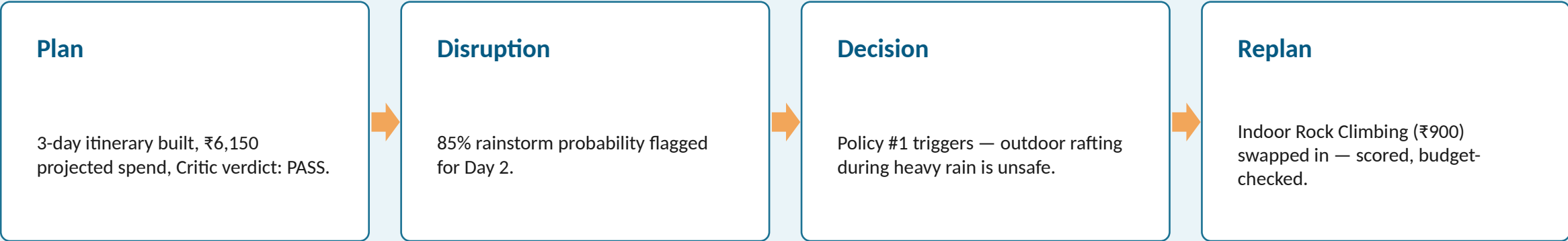
Agent	Module	Core Function
Planner	<i>planner.py</i>	Synthesizes trip goals into initial day-by-day itineraries.
Weather	<i>weather.py</i>	Analyzes forecasts against scheduled outdoor activities.
Mobility	<i>mobility.py</i>	Calculates inter-activity distances, transit durations, delays.
Venue	<i>venue.py</i>	Monitors opening hours and discovers alternatives.
Budget	<i>budget.py</i>	Tracks spend and evaluates replacement cost viability.
Critic	<i>critic.py</i>	Runs the 5-dimension audit (Budget, Weather, Travel, Venues, Preferences).
Monitoring	<i>monitor.py</i>	Surveys signals and aggregates active disruption events.
Replanner	<i>replanner.py</i>	Executes targeted substitution, preserving unaffected slots.

The Agentic Workflow — closed control loop



DEMONSTRATION SCENARIO

"Plan a 3-day trip to Rishikesh for ₹15,000 with adventure activities and good food."



Minimal replanning — proved

Day 1	PRESERVED — 100% untouched
Day 2 AM	REPLACED — rafting → indoor climbing (saves ₹600)
Day 2 PM/Eve	PRESERVED — untouched
Day 3	PRESERVED — 100% untouched

Why it matters

- Zero hallucinated finances — arithmetic is deterministic code, not LLM guesswork.
- Safety-first — high-risk activities are auto-restricted under severe weather.
- Explainable — every change ships with a plain-language rationale.
- Resilient — automatic fallback to certified simulations if live APIs fail.

Roadmap & Fair Play

What's next

- Multi-modal transit optimization with live public transit feeds.
- Group Consensus Agent for multi-traveler preference reconciliation.
- Live flight-disruption sync via airline tracking APIs.

Built responsibly

- No personal credentials or private travel documents collected or logged.
- Deterministic safety policies over high-risk outdoor activities.
- Full audit trail for every autonomous decision.

Team Versetify

VoyageOS — Autonomous Travel Planning & Re-Planning Agent

Repository

github.com/Priyank-14/voyage-os

Event

Agentic AI Hackathon, Tech Zephyr 4.0 — IIT Bhubaneswar

Tech stack

Python 3.11+ · Pydantic v2 · Streamlit · SQLite · Pytest

Thank you — questions welcome.